

Naive Bayes Foundations: A Bayes' Theorem Problem

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Conditional probability ([Note MA-015](#)); Joint and marginal probability ([Note MA-014](#)).
- **Leads to:** Naive Bayes ([Note ML-081](#)); MAP estimation ([Note MA-072](#)); Gaussian mixture model (GMM) ([Note MA-073](#)).

1. Overview

Key point: Three machines make markers; a random marker turns out defective. Bayes' theorem, with the total probability rule for the evidence, gives the chance it came from each machine.

This Note works through a classic problem with Bayes' theorem. The Note also introduces the **law of total probability** (G-1053), the usual way to compute the evidence $P(B)$ in the denominator (the overall probability of what we observed, here a defective marker). Naive Bayes computes its evidence the same way.

2. The problem

Key point: Production shares 20%, 30%, 50%; defect rates 5%, 3%, 1%.

A factory makes markers on three machines:

Machine	Share of all markers	Defect rate of that machine
M1	20%	5%
M2	30%	3%
M3	50%	1%

A marker is picked at random from a batch and turns out to be **defective**. What is the probability that it was made by M3?

Figure 1 shows the idea before any formula. In a typical batch of 1000 markers, M1, M2 and M3 make 200, 300 and 500. Their defect rates leave 10, 9 and 5 defective markers. Knowing the picked marker is defective means it is one of these 24, and only 5 of the 24 came from M3.

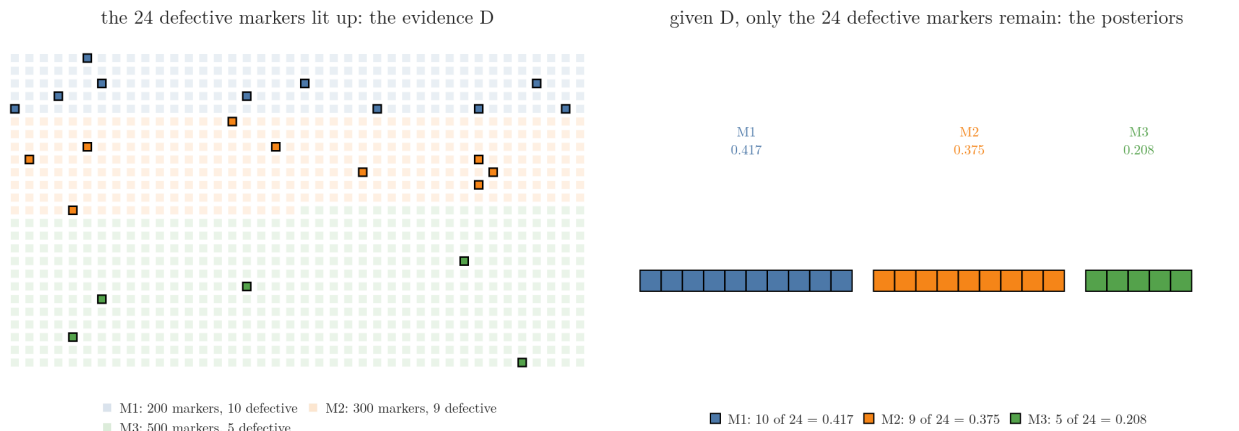


Figure 1: A batch of 1000 markers, one square each, coloured by machine. The 24 defective ones light up (10 from M1, 9 from M2, 5 from M3); given that the marker is defective, only those 24 remain, and M3 supplies 5 of 24 = 0.208.

3. Writing down what we know

Key point: The production shares are priors $P(M)$; the defect rates are likelihoods $P(D | M)$.

Let D be “the marker is defective”. In probability language:

- **Priors** (G-1565): $P(M1) = 0.2$, $P(M2) = 0.3$, $P(M3) = 0.5$.
- **Likelihoods** (G-1086): $P(D | M1) = 0.05$, $P(D | M2) = 0.03$, $P(D | M3) = 0.01$.

The question asks for $P(M3 | D)$. By **Bayes’ theorem** (G-269; it turns $P(D | M3)$, the chance of a defect given the machine, into $P(M3 | D)$, the chance of the machine given a defect; see [the names of the four parts](#)),

$$P(M3 | D) = \frac{P(D | M3) \times P(M3)}{P(D)}$$

Everything on the right is known except $P(D)$, the overall probability that a marker is defective.

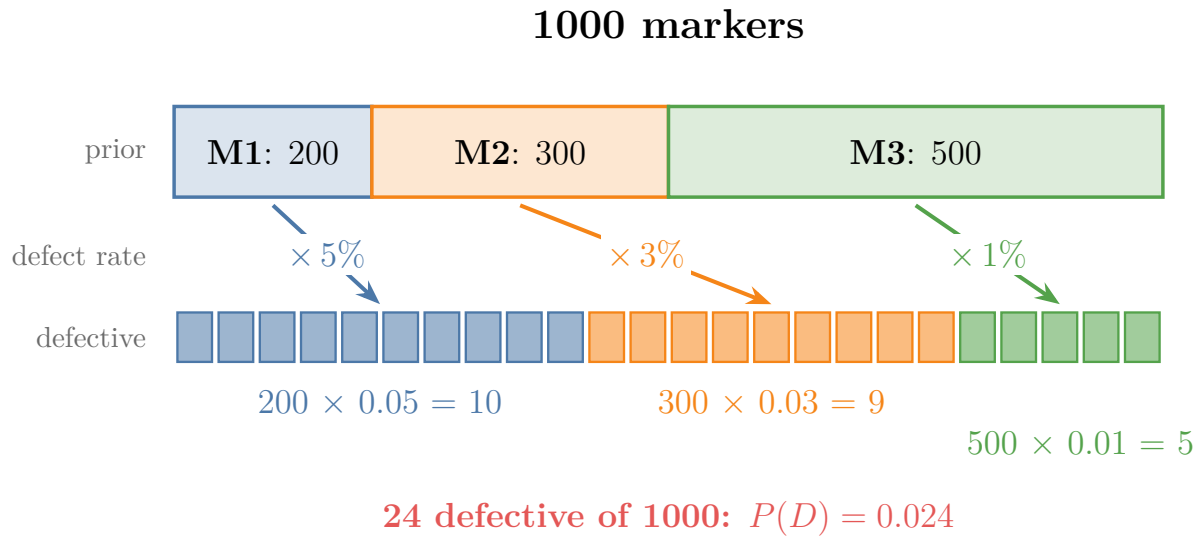


Figure 2: The problem as counts in a batch of 1000 markers. Top: the priors, with widths to scale. Bottom: one square per defective marker, from each machine’s count times its defect rate

Figure 2 turns the priors and likelihoods into expected counts for 1000 markers. Watch the bottom row: M3 makes the most markers but supplies the fewest defective squares.

4. The evidence: total probability

Key point: A defective marker came from exactly one machine, so $P(D)$ is the sum of the three “defective and from machine i ” probabilities.

A defective marker must have come from M1, M2 or M3, and from only one of them. These three cases are **mutually exclusive** (they cannot happen together), so their probabilities add. The symbol $\cap$ means “and”: $D \cap M1$ is “defective and made by M1”.

$$\begin{aligned}
 P(D) &= P(D \cap M1) \\
 &\quad + P(D \cap M2) \\
 &\quad + P(D \cap M3)
 \end{aligned}$$

Each term is a likelihood times a prior, by this rule from [the proof of Bayes’ theorem](#):

$$P(A \cap B) = P(B | A) P(A)$$

So for every machine,

$$P(D \cap M) = P(D \mid M) P(M)$$

and the sum becomes

$$\begin{aligned} P(D) &= P(D \mid M1)P(M1) \\ &+ P(D \mid M2)P(M2) \\ &+ P(D \mid M3)P(M3) \end{aligned}$$

This sum is the law of total probability. With numbers:

$$\begin{aligned} P(D) &= 0.05 \times 0.2 \\ &+ 0.03 \times 0.3 \\ &+ 0.01 \times 0.5 \end{aligned}$$

$$P(D) = 0.010 + 0.009 + 0.005$$

$$P(D) = 0.024$$

So 2.4% of all markers are defective. Figure 3 shows the same calculation as a **probability tree** (G-1573). Each path from the left to a defective marker passes through one machine, and multiplying the numbers along the path gives that machine's term of the sum above.

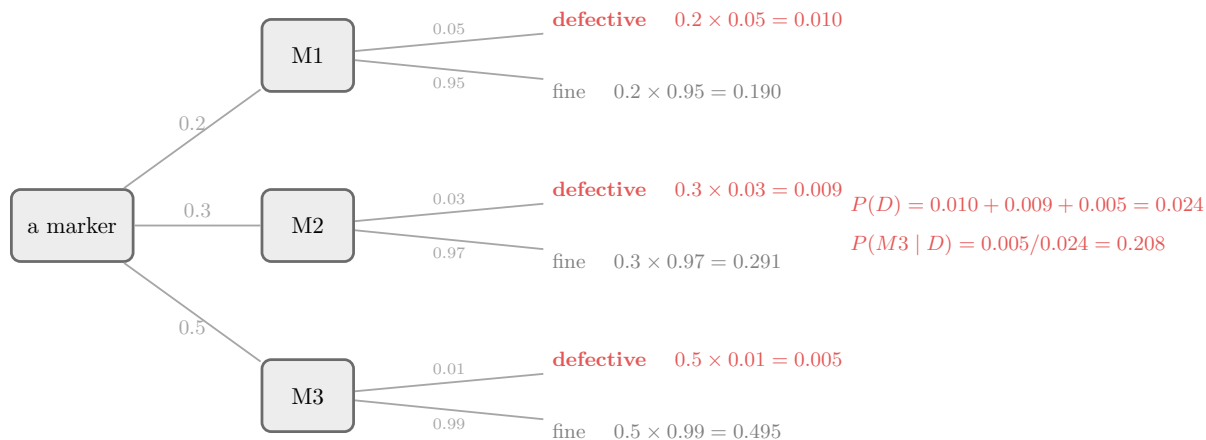


Figure 3: Probability tree: each path multiplies its branches

5. The answer

Key point: Given a defect, the chance that M3 made the marker is only 0.208: M3 makes half the markers but only about a fifth of the defective ones.

$$\begin{aligned}
 P(M3 \mid D) &= \frac{0.01 \times 0.5}{0.024} \\
 &= \frac{0.005}{0.024} \\
 &= 0.208
 \end{aligned}$$

The same calculation for every machine:

Machine	Prior $P(M)$	Joint probability (G-986) $P(D \cap M)$	Posterior (G-1536) $P(M \mid D)$
M1	0.20	0.010	0.417
M2	0.30	0.009	0.375
M3	0.50	0.005	0.208

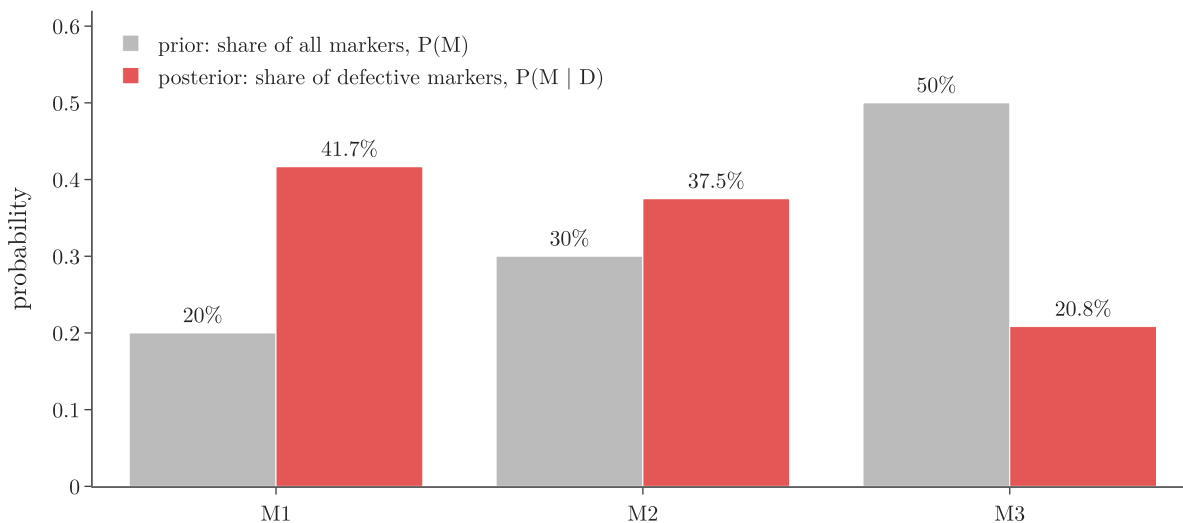


Figure 4: Share of all markers against share of defective markers

In Figure 4, each machine has two bars: its share of all markers and its share of the defective ones. Seeing that the marker is defective changes the picture completely. Before, M3 was the most likely source (50%). After, M3 is the least likely (21%), because M3 rarely makes defects.

An everyday picture: three cooks share a kitchen, and one dish comes out burnt. The cook who makes the most dishes is not the likely culprit if that cook almost never burns anything. M1 makes only 20% of markers but 42% of the defective ones.

The three posteriors add up to 1, as they must: the marker came from some machine. A simulation of a million markers gives $P(M3 | D) = 0.2095$, matching 0.208.

Python: The whole calculation.

```
prior = {"M1": 0.2, "M2": 0.3, "M3": 0.5}
p_defect = {"M1": 0.05, "M2": 0.03, "M3": 0.01}

joint = {m: prior[m] * p_defect[m] for m in prior}
p_D = sum(joint.values()) # 0.024
posterior = {m: joint[m] / p_D for m in prior} # 0.417, 0.375, 0.208
```

6. Another way to see it: a medical test

Key point: A test does not give the final answer; it updates the prior. Posterior odds = prior odds $\times$ Bayes factor.

The marker problem has the same shape as a famous question about medical tests. The machines become “has cancer” and “does not have cancer”; “defective” becomes “the test is positive”.

A screening test is given to 1000 women. What we know:

- **Prior:** 1 percent have breast cancer, so 10 women are sick and 990 are healthy.
- **Sensitivity** (G-1641; also called recall, [the share of real positives found](#)): the test is positive for 90 percent of the sick women: $P(+ | \text{cancer}) = 0.9$.
- **False positive rate** (G-749): the test is also positive for 9 percent of the healthy women: $P(+ | \text{healthy}) = 0.09$.

A woman tests positive. What is the chance that she has cancer? The test sounds “90 percent accurate”, so many people answer 9 in 10. Count instead, exactly as with the markers:

1. **Sick and positive** (true positives):

$$10 \times 0.9 = 9 \text{ women}$$

2. **Healthy and positive** (false positives):

$$990 \times 0.09 = 89.1 \approx 89 \text{ women}$$

3. **All positives** (the evidence, by total probability):

$$9 + 89 = 98$$

4. **Posterior:** the share of positives who are sick.

$$P(\text{cancer} \mid +) = 9/98 \approx 0.092$$

That is about 1 in 11.

The healthy group is so large that its few false positives (89) swamp the true positives (9). This posterior is the **precision** (G-1547) of the test; in medicine it is called the positive predictive value.

The lesson: the test did not decide anything. The test moved the chance from 1 in 100 (the prior) to about 1 in 11 (the posterior). A test **updates** the prior.

6.1 The size of the update: the Bayes factor

Key point: Bayes factor = $P(\text{evidence} \mid \text{yes}) / P(\text{evidence} \mid \text{no})$. Multiply the prior odds by it to get the posterior odds.

How strong is the update? A positive result is 10 times more likely for a sick woman than for a healthy one:

$$0.9/0.09 = 10$$

This ratio of the two likelihoods is the **Bayes factor** (G-2215).

The Bayes factor works on **odds** (G-1376): the number of “yes” cases to the number of “no” cases.

1. **Prior odds:** 10 sick to 990 healthy, that is 1 : 99.
2. **Multiply by the Bayes factor.** These are the posterior odds.

$$1 \times 10 : 99 = 10 : 99$$

3. **Back to a probability:** 10 sick for every 99 healthy.

$$10/(10 + 99) = 0.092$$

The same answer as the count.

The rule is exact, and it follows from the count above: among the positives, the sick are scaled by 0.9 and the healthy by 0.09, so their ratio is multiplied by 0.9/0.09.

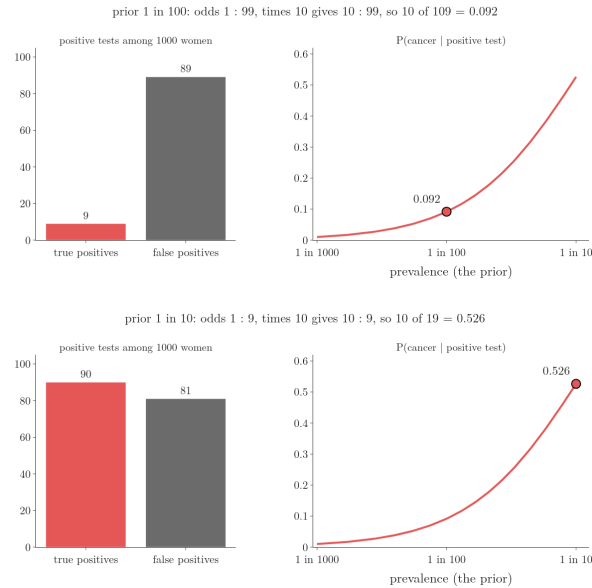


Figure 5: A test with sensitivity 0.9 and false positive rate 0.09 (Bayes factor 10), for priors from 1 in 1000 to 1 in 10. Left: expected true and false positives among 1000 women. Right: the posterior $P(\text{cancer} \mid \text{positive})$. The title gives the update in odds.

In Figure 5 the horizontal axis of the right panel is a log scale: 1 in 1000, 1 in 100 and 1 in 10 are equally spaced, each ten times the one before. Watch the same test give very different answers as the prior changes. At a prior of 1 in 1000 a positive result means about 1 percent; at 1 in 100, 9 percent; at 1 in 10, the odds 1 : 9 become 10 : 9, which is 53 percent. The test is the same each time; only the prior differs.

The Bayes factor also previews Naive Bayes: each feature multiplies the score of a class by its own likelihood, one factor per feature ([the naive assumption](#)).

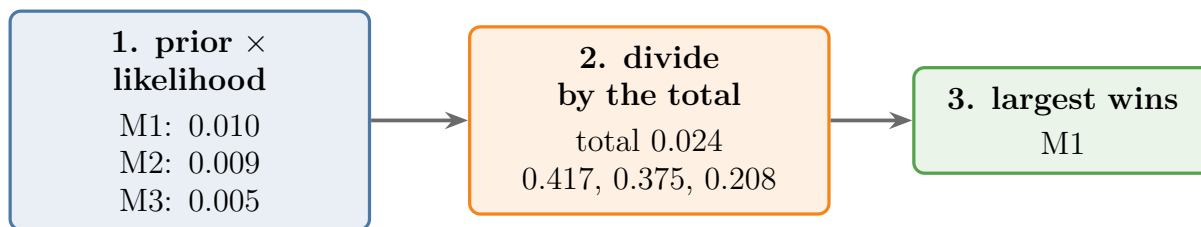
7. The pattern for Naive Bayes

Key point: For each class: prior $\times$ likelihood. Divide by their total. The largest posterior wins.

A Naive Bayes classifier follows exactly this procedure. The machines become the classes, the values of the **target** (G-1949; the output we predict). “Defective” becomes the observed **features** (G-772; the input variables, one column each of the data table):

1. for every class, multiply its prior by the likelihood of the observed features;
2. divide each product by their sum (the evidence) to get posteriors that add to 1;
3. predict the class with the largest posterior.

Since the evidence is the same for every class, step 2 does not change which class wins; it only turns the scores into probabilities.



step 2 rescales every score by the same number: the winner does not change

Figure 6: The three Naive Bayes steps on the marker problem

Figure 6 runs the three steps on the marker numbers: M1 has the largest score both before and after dividing by 0.024.

8. Summary

- Priors: production shares; likelihoods: defect rates; posterior: which machine, given a defect.
- Law of total probability: $P(D) = \sum_i P(D | M_i)P(M_i)$. The sign $\sum_i$ means “add the terms for $i = 1, 2, 3$ ”, one per machine (0.010, 0.009 and 0.005, which add to 0.024).
- Answer: $P(M3 | D) = 0.208$; M1 is the most likely source (0.417).
- Medical test: prior 1 in 100, sensitivity 0.9, false positive rate 0.09 give a posterior of only 0.092 (9 sick among 98 positives). Posterior odds = prior odds $\times$ Bayes factor (section 6.1).

9. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 5 | Problem based upon Bayes Theorem”, YouTube, <https://www.youtube.com/watch?v=aAEHjXDHtbE>
- Sanderson, G. (3Blue1Brown), “The medical test paradox, and redesigning Bayes’ rule”, YouTube, <https://www.youtube.com/watch?v=lG4VkPoG3ko> (the 1000-women example, the test as an update, the Bayes factor and the odds form; Figure 5 is our own chart of those numbers)

10. Key terms

Term	Meaning
Law of total probability	$P(B) = \sum_i P(B A_i)P(A_i)$, when the A_i are mutually exclusive and cover every case
Probability tree	A diagram in which each path multiplies the probabilities along its branches
Bayes factor (G-2215)	The ratio of the two likelihoods, $P(\text{evidence} \text{yes})/P(\text{evidence} \text{no})$; it multiplies the prior odds to give the posterior odds
Odds	The number of “yes” cases to the number of “no” cases, such as 1 : 99
False positive rate (G-749)	The share of real negatives that test positive
Joint probability	The probability that two events happen together, $P(A \cap B)$